ONLINE SURVEY SYSTEM

Java Console-Based Mini Project – Source Code and Execution Guide

Features: Admin login, create surveys, view available surveys, submit respondent answers, and view vote counts with percentages.

Demo login: Username: admin Password: 1234

Note: This beginner-friendly version stores data in memory. Surveys and responses are cleared when the program closes. It is a console-based prototype, not a deployed multi-user website. A real online version needs a web framework and database.

1. Java Source Code

```
import java.util.ArrayList;
import java.util.Scanner;

public class OnlineSurveySystem {
    static Scanner sc = new Scanner(System.in);
    static ArrayList<Survey> surveys = new ArrayList<>();

    static class Survey {
        int id;
        String title;
        String question;
        String[] options;
        int[] votes;

        Survey(int id, String title, String question, String[] options) {
            this.id = id;
            this.title = title;
            this.question = question;
            this.options = options;
            this.votes = new int[options.length];
        }
    }

    public static void main(String[] args) {
        int choice;
        do {
            System.out.println("\n===== ONLINE SURVEY SYSTEM =====");
            System.out.println("1. Admin Login");
            System.out.println("2. Respondent");
            System.out.println("3. Exit");
            System.out.print("Enter choice: ");
            choice = readInt();

            switch (choice) {
                case 1:
                    adminLogin();
                    break;
                case 2:
                    respondentMenu();
                    break;
                case 3:
                    System.out.println("Thank you for using the system!");
                    break;
                default:
                    System.out.println("Invalid choice.");
            }
        } while (choice != 3);
    }

    static void adminLogin() {
        System.out.print("Enter admin username: ");
        String username = sc.nextLine();
        System.out.print("Enter admin password: ");
        String password = sc.nextLine();
```

ONLINE SURVEY SYSTEM – JAVA SOURCE CODE (continued)

```
        // Demo credentials. Change these before real deployment.
        if (username.equals("admin") && password.equals("1234")) {
            adminMenu();
        } else {
            System.out.println("Invalid username or password.");
        }
    }

    static void adminMenu() {
        int choice;
        do {
            System.out.println("\n----- ADMIN MENU -----");
            System.out.println("1. Create Survey");
            System.out.println("2. View Surveys");
            System.out.println("3. View Survey Results");
            System.out.println("4. Back");
            System.out.print("Enter choice: ");
            choice = readInt();

            switch (choice) {
                case 1:
                    createSurvey();
                    break;
                case 2:
                    viewSurveys();
                    break;
                case 3:
                    viewResults();
                    break;
                case 4:
                    break;
                default:
                    System.out.println("Invalid choice.");
            }
        } while (choice != 4);
    }

    static void createSurvey() {
        System.out.print("Enter survey title: ");
        String title = sc.nextLine();
        System.out.print("Enter survey question: ");
        String question = sc.nextLine();

        int count;
        do {
            System.out.print("Enter number of options (2 to 5): ");
            count = readInt();
        } while (count < 2 || count > 5);

        String[] options = new String[count];
        for (int i = 0; i < count; i++) {
            System.out.print("Enter option " + (i + 1) + ": ");
            options[i] = sc.nextLine();
        }
```

ONLINE SURVEY SYSTEM – JAVA SOURCE CODE (continued)

```
        int id = surveys.size() + 1;
        surveys.add(new Survey(id, title, question, options));
        System.out.println("Survey created successfully. Survey ID: " + id);
    }

    static void viewSurveys() {
        if (surveys.isEmpty()) {
            System.out.println("No surveys available.");
            return;
        }

        System.out.println("\n----- AVAILABLE SURVEYS -----");
        for (Survey s : surveys) {
            System.out.println("ID: " + s.id + " | Title: " + s.title);
            System.out.println("Question: " + s.question);
        }
    }

    static void respondentMenu() {
        if (surveys.isEmpty()) {
            System.out.println("No surveys are available at present.");
            return;
        }

        viewSurveys();
        System.out.print("\nEnter the survey ID you want to answer: ");
        int id = readInt();
        Survey selected = findSurvey(id);

        if (selected == null) {
            System.out.println("Survey not found.");
            return;
        }

        System.out.println("\n" + selected.title);
        System.out.println(selected.question);
        for (int i = 0; i < selected.options.length; i++) {
            System.out.println((i + 1) + ". " + selected.options[i]);
        }

        System.out.print("Select your answer: ");
        int answer = readInt();
        if (answer >= 1 && answer <= selected.options.length) {
            selected.votes[answer - 1]++;
            System.out.println("Your response has been submitted.");
        } else {
            System.out.println("Invalid option. Response not submitted.");
        }
    }

    static void viewResults() {
        if (surveys.isEmpty()) {
            System.out.println("No survey results available.");
            return;
        }
```

ONLINE SURVEY SYSTEM – JAVA SOURCE CODE (continued)

```
        viewSurveys();
        System.out.print("\nEnter survey ID to view results: ");
        int id = readInt();
        Survey s = findSurvey(id);

        if (s == null) {
            System.out.println("Survey not found.");
            return;
        }

        int total = 0;
        for (int vote : s.votes) {
            total += vote;
        }

        System.out.println("\n----- SURVEY ANALYSIS -----");
        System.out.println("Survey: " + s.title);
        System.out.println("Question: " + s.question);
        System.out.println("Total responses: " + total);

        for (int i = 0; i < s.options.length; i++) {
            double percentage = total == 0 ? 0 :
                    (s.votes[i] * 100.0 / total);
            System.out.printf("%s : %d votes (%.1f%%)%n",
                    s.options[i], s.votes[i], percentage);
        }
    }

    static Survey findSurvey(int id) {
        for (Survey s : surveys) {
            if (s.id == id) {
                return s;
            }
        }
        return null;
    }

    static int readInt() {
        while (!sc.hasNextInt()) {
            System.out.print("Please enter a valid number: ");
            sc.next();
        }
        int value = sc.nextInt();
        sc.nextLine(); // Clear the input buffer.
        return value;
    }
}
```

2. How to Execute the Program

1. Install the Java Development Kit (JDK) on your computer.

2. Open Notepad, VS Code, or any Java editor.

3. Copy the complete source code from this PDF.

4. Save the file with the exact name **OnlineSurveySystem.java**.

5. Open Command Prompt or Terminal in the folder where the file is saved.

6. Compile the program using: **javac OnlineSurveySystem.java**.

7. Run the program using: **java OnlineSurveySystem**.

8. Choose **1** for Admin Login and enter username **admin** and password **1234**.

9. Create a survey by entering its title, question, and 2–5 options.

10. Return to the main menu and choose **2** to answer a survey.

11. Log in as admin again and choose **3** to view the survey results.

3. Main Modules

• **Admin:** Authenticates the administrator and manages surveys.

• **Survey Creator:** Creates a survey with a title, question, and multiple-choice options.

• **Respondent:** Displays surveys and records the selected answer.

• **Analysis:** Calculates total responses, option-wise votes, and percentages.

Important: This is suitable as a basic Java mini-project demonstration. For a genuinely online system with multiple users and permanent records, extend it using Java Servlets/JSP or Spring Boot with a MySQL database, user authentication, and web pages.